

Software Engineering

Banking System Report

A: Risk identification

Risk ID	Risks Description	Risk type
Risk 1	Illegal access to the bank system and the reach to sensitive data for bank system users will lead to data leak and customer's lack of trust.	Technology risk
Risk 2	Change of economy or politics and not being able to track technological evolution would lead to the loss of customers.	Organisational risk
Risk 3	Failure of marketing plans where marketing is one of the best ways to publish a bank system and grab as many users as possible.	People risk
Risk 4	Taking long time to finish the project leads to change in goals and have less resources.	Estimation risk
Risk 5	Older systems may cause unexpected errors.	Technology risk
Risk 6	The project itself may undergo sudden changes, such as changing the definition of the project or the idea.	Requirements risk
Risk 7	Requirements set by manager or client are not clear.	Requirements risk
Risk 8	Lack of skilled workers, this can lead to big problems, especially with cybersecurity, following rules, and understanding new money technology.	People risk
Risk 9	Relying on inaccurate business tools such as excel may lead to inaccurate decisions regarding employees.	Tools risk
Risk 10	After launching, problems may occur in the daily operation of the system such as system crashes or slow performance.	Technology risk
Risk 11	After launching, users may leave their accounts on run without logging out which leads to data breach.	Technology risk

B: Risk analysis

Risk ID	Probability	Effects
Risk 1	MODERATE	Catastrophic
Risk 2	MODERATE	Tolerable
Risk 3	LOW	Tolerable
Risk 4	MODERATE	Serious
Risk 5	MODERATE	Serious
Risk 6	HIGH	Serious
Risk 7	HIGH	Serious
Risk 8	MODERATE	Serious
Risk 9	MODERATE	Catastrophic
Risk 10	HIGH	Serious
Risk 11	HIGH	Serious

C: Risk mitigation

Risk ID	Strategy
Risk 1	While developing the system, developers should implement security strategies, encrypt sensitive data and MAF- multi authentication factor.
Risk 2	The team should continuously track the economic evolution to keep up with any changes that occur.
Risk 3	The marketing team should study their plans very well and use the best ways to advertise the system so that they could grab as much users as possible.
Risk 4	Hire more efficient staff to make the team go faster and better.
Risk 5	Conduct several quality tests such as unit testing and try to create a specialised team for software errors so that it is ready to deal with errors as soon as they are discovered.
Risk 6	The team must always have alternative plans and strategies.
Risk 7	The team asks experts to clarify these unclear requirements.
Risk 8	Training courses to improve the employee's skill in cybersecurity.
Risk 9	The team evaluates and verifies the accuracy of the tools.
Risk 10	We, as a team, must monitor and analyse the system periodically and develop a plan for continuous improvement of the system and conduct periodic maintenance. The system must also be designed with a system recovery plan and backup in the event of a failure.
Risk 11	Implementing technical solutions to minimize risk such as implement auto log out.

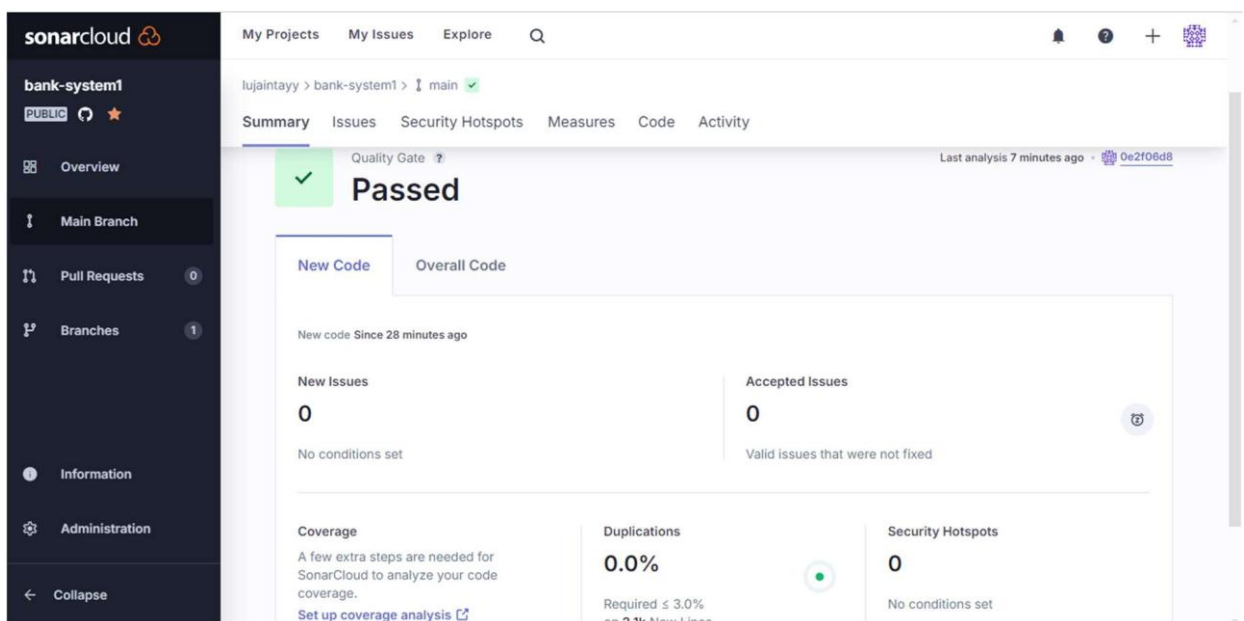
Risk ID	Risk type	Potential indicator
Risk 1	Technology risk	Weak security procedures; malware attacks, viruses and phishing.
Risk 2	Organisational risk	Rapid economic evolution; not following the technological and economic changes.
Risk 3	People risk	Using wrong marketing tools; lack of skilled marketing team.
Risk 4	Estimation risk	Lack of skilled staff and resources; teamwork absence.
Risk 5	Technology risk	Ignoring quality test such as unit test; systems update.
Risk 6	Requirements risk	Not taking precautions for any sudden change, requirements changes.
Risk 7	Requirements risk	Weak communication between manager and team members.
Risk 8	People risk	Shortage of skilled cybersecurity employees in labour market.
Risk 9	Tools risk	Technical issues.
Risk 10	Technology risk	Continuous maintenance.
Risk 11	Technology risk	Developer disregard of details.

A: Code Quality Analyses:

- We tested our code quality using GitHub with SonarCloud tool, as the report shows below, our code passed with no issues or improvements needed.

[GITUHB LINK](#)

Report:



B: Reliability Measurement:

 Ran today at 23:48:15 · [View all runs](#)

Source	Environment	Iterations	Duration	All tests	Avg. Resp. Time
Runner	none	200	19s 779ms	1000	82 ms

[All Tests](#) Passed (998) Failed (2) Skipped (0) [View S...](#)

1. Probability of failure on demand (PFD):

- $PFD = \text{Failed Test} / \text{Total Tests}$
- Failed = 2, Total = 1000
- $PFD = 2 / 1000 = 0.002$ or 0.2%.

2. Rate of occurrence of failures (ROCOF):

- $ROCOF = \text{Failed Test} \setminus \text{Duration}$
- Failed = 2, Duration = 19.779 seconds
- $ROCOF = 2 / 19.779 = 0.101$ failures per second.

3. Mean Time Between Failures (MTBF):

- $MTBF = \text{Total Duration} \setminus \text{Number of Failures}$
- Total Duration = 19.779 seconds, Number of Failures = 2
- $MTBF = 19.779 \setminus 2 = 9.89$ seconds between failures.

4. Availability (A):

- $A = \text{Passed Tests} \setminus \text{Total Tests}$
- Passed = 998, Total = 1000
- $A = 998 \setminus 1000 = 0.998$ or 99.8%.

C: Security Analyses:

1- Security Method (code):

```
1 import javafx.application.Application;
2 import javafx.scene.Scene;
3 import javafx.scene.control.Button;
4 import javafx.scene.control.Label;
5 import javafx.scene.layout.*;
6 import javafx.stage.Stage;
7
8 public class QuickAccessApp extends Application {
9
10     @Override
11     public void start(Stage primaryStage) {
12         VBox root = new VBox(24);
13         root.setPadding(new javafx.geometry.Insets(24));
14         root.setStyle("-fx-background-color: #F5F5F5;");
15
16         root.getChildren().addAll(
17             new Label("Welcome back!") {{
18                 setStyle("-fx-font-size: 24px; -fx-font-weight: bold;");
19             }},
20             new Label("Enter your PIN to access your account"),
21             createPinDots(),
22             createButtonPanel(),
23             new Label("Forgot PIN?") {{
24                 setStyle("-fx-text-fill: #6200EE;");
25             }}
26         );
27
28         primaryStage.setScene(new Scene(root, 400, 400));
29         primaryStage.setTitle("Quick Access");
30         primaryStage.show();
31     }
32
33     private HBox createPinDots() {
34         HBox dots = new HBox(16);
35         for (int i = 0; i < 3; i++) {
36             dots.getChildren().add(new Region() {{
37                 setPrefSize(12, 12);
38                 setStyle("-fx-background-color: #6200EE; -fx-shape: 'M0,0 L12,0 L12,12 L0,12 Z'");
39             }});
40         }
41         return dots;
42     }
43
44     private HBox createButtonPanel() {
45         HBox buttons = new HBox(16);
46         buttons.getChildren().addAll(
47             createButton("1"),
48             new Region(),
49             createButton("0"),
50             new Button("-") {{
51                 setOnAction(e -> System.out.println("Backspace pressed ..."));
52             }}
53         );
54         return buttons;
55     }
56
57     private Button createButton(String text) {
58         return new Button(text) {{
59             setPrefSize(80, 80);
60             setStyle("-fx-background-color: #FFFFFF; -fx-border-color: #6200EE; -fx-border-radius: 40;");
61             setOnAction(e -> System.out.println(text + " button pressed ..."));
62         }};
63     }
64
65     public static void main(String[] args) {
66         launch(args);
67     }
68 }
```


2- Asset analysis:

Asset	Value	Exposure
Customer Database	Very high Contains personal and confidential information	High Stealing customers confidential information This leads to loss of customer trust.
Core Banking System	High Bank's main operations and it is necessary to provide services.	High May be exposed to cyber attacks that may compromise sensitive data.

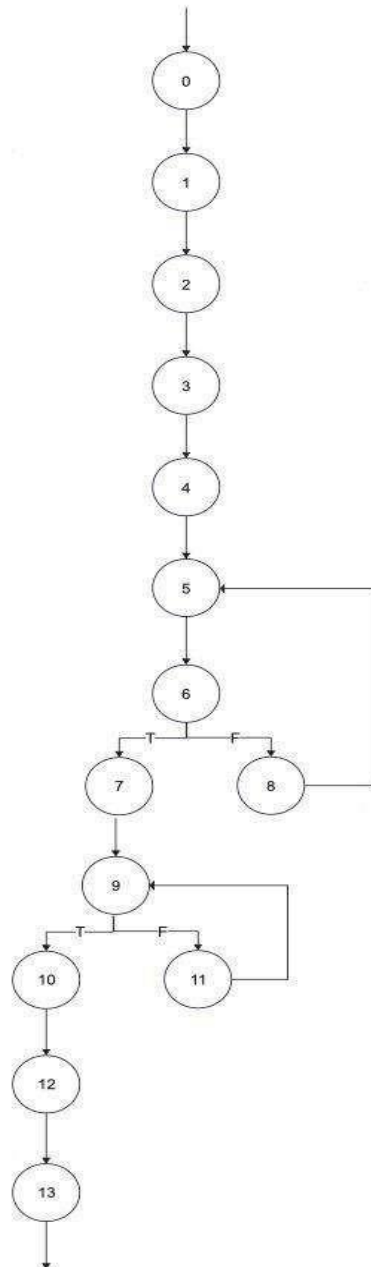
Threat	Probability	Control	Feasibility
Unauthorized Access to Systems	High	Only allow system management from specific locations that are physically secure.	high-cost solution. Possible user resistance.
Data Breaches	High	Encrypt sensitive data	Strong measures can be implemented.

```

1 import javafx.application.Application;
2 import javafx.scene.Scene;
3 import javafx.scene.control.Button;
4 import javafx.scene.control.Label;
5 import javafx.scene.layout.*;
6 import javafx.stage.Stage;
7
8 public class QuickAccessApp extends Application {
9
10     @Override
11     public void start(Stage primaryStage) {
12         VBox root = new VBox(24);
13         root.setPadding(new Insets(24));
14         root.setStyle("-fx-background-color: #F5F5F5;");
15
16         root.getChildren().addAll(
17             new Label("Welcome back!") {
18                 setStyle("-fx-font-size: 24px; -fx-font-weight: bold;");
19             },
20             new Label("Enter your PIN to access your account"),
21             createPinDots(),
22             createButtonPanel(),
23             new Label("Forgot PIN?") {
24                 setStyle("-fx-text-fill: #6200EE;");
25             }
26         );
27
28         primaryStage.setScene(new Scene(root, 480, 480));
29         primaryStage.setTitle("Quick Access");
30         primaryStage.show();
31     }
32
33     private HBox createPinDots() {
34         HBox dots = new HBox(16);
35         for (int i = 0; i < 3; i++) {
36             dots.getChildren().add(new Region() {
37                 setPrefSize(12, 12);
38                 setStyle("-fx-background-color: #6200EE; -fx-shape: 'M0,0 L12,0 L12,12 L0,12 Z'");
39             });
40         }
41         return dots;
42     }
43
44     private HBox createButtonPanel() {
45         HBox buttons = new HBox(16);
46         buttons.getChildren().addAll(
47             createButton("1"),
48             new Region(),
49             createButton("0"),
50             new Button("<") {
51                 setOnAction(e -> System.out.println("Backspace pressed ..."));
52             }
53         );
54         return buttons;
55     }
56
57     private Button createButton(String text) {
58         return new Button(text) {
59             setPrefSize(80, 80);
60             setStyle("-fx-background-color: #FFFFFF; -fx-border-color: #6200EE; -fx-border-radius: 40;");
61             setOnAction(e -> System.out.println(text + " button pressed ..."));
62         };
63     }
64
65     public static void main(String[] args) {
66         launch(args);
67     }
68 }

```

2. Control Flow Graph:



3. Cyclomatic complexity:

Cyclomatic Complexity:

Region = 3 $\rightarrow$ cyclomatic complexity = 3

Predicates = 2 $\rightarrow$ cyclomatic complexity = 2+1 = 3

Nodes = 14
Edges = 15 $\rightarrow$ cyclomatic complexity = $E - N + 2$
 $= 15 - 14 + 2$
 $= 1 + 2$
 $= 3$

Basic paths:

①: [0-1-2-3-4-5-6-7-9-10-12-13]

②: [0-1-2-3-4-5-6-8-5-6-7-9-10-12-13]

③: [0-1-2-3-4-5-6-7-9-11-9-10-12-13]

Derive Test Cases:

Path: [0-1-2-3-4-5-6-7-9-10-12-13]

test case: value of $i = 2$
 $2 < 3 \rightarrow T$
user choose (1)
 $1 \rightarrow \text{yes}$

Expected output:

Go to "Homepage"

Path: [0-1-2-3-4-5-6-8-5-6-7-9-10-12-13]

test case: value of $i = 4$
 $4 > 3 \rightarrow F$
user choose (1)
 $1 \rightarrow \text{yes}$

Expected output:

$\rightarrow$ Re-enter value of i
 $\rightarrow$ Go to "Homepage"

Path: [0-1-2-3-4-5-6-7-9-11-9-10-12-13]

test case: value of $i = 1$
 $1 < 3 \rightarrow \text{True}$
user choose (0)

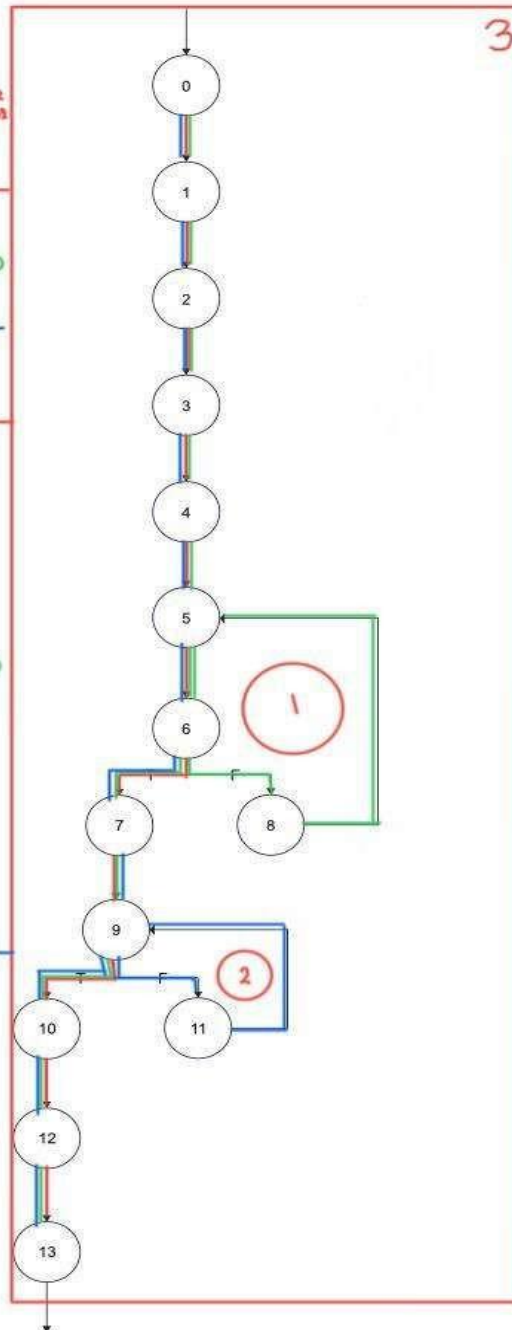
$0 \rightarrow$ " backpage "

Expected output:

$\rightarrow$ Go back to " backpage "

$\rightarrow$ User choose (1)

$\rightarrow$ Go to homepage



B: Black-box testing

Requirement 1: The user shall be able to log in through their ID number and password.

Test Case 1.1	Description	Preconditions	Test Steps	Expected Result
Valid Login (Equivalence Partitioning)	Verify that the user can log in with a valid ID number and password.	User has a valid account.	1. Navigate to the login page. 2. Enter a valid ID number 3. Enter a valid password 4. Click the "Login" button.	User is logged in successfully, and the account dashboard is displayed.

Test Case 1.2	Description	Preconditions	Test Steps	Expected Result
Invalid Login (Equivalence Partitioning)	Verify that the user cannot log in with an invalid ID number or password.	User does not have an account with the given credentials.	1. Navigate to the login page. 2. Enter an invalid ID number . 3. Enter an invalid password 4. Click the "Login" button.	Login fails, and the system displays an error message: "Invalid ID or password."

Requirement 2: The user shall be able to check their account balance once logged in to their account.

Test Case 2.1	Description	Preconditions	Test Steps	Expected Result
Check Account Balance After Login (Equivalence Partitioning)	Verify that the user can check their account balance after a successful login	The user has a registered account and is logged in.	1. Log in using valid credentials 2. Navigate to the account balance section.	The system displays the correct account balance.

Test Case 2.2	Description	Preconditions	Test Steps	Expected Result
Attempt to Check Balance Without Login (Equivalence Partitioning)	Verify that the user cannot check their account balance without logging in.	The user is not logged into the system. □	Without logging in, attempt to access the "Account Balance" section	Access is denied, and the system displays a message: "Please log in to view your account balance."

Requirement 3: The user shall be able to create several accounts on the same device.

Test Case 3.1	Description	Preconditions	Test Steps	Expected Result
Create Multiple Accounts Successfully (Equivalence Partitioning)	Verify that the user can create multiple accounts on the same device using different credentials.	No restrictions on creating multiple accounts on the same device.	1. Create a new account with ID , Password: 2. Log out of the first account. 3. Create a second account with a different ID, Password 4. Log into each account one by one.	Both accounts are created successfully, and the user can log in to each account individually.

Test Case 3.2	Description	Preconditions	Test Steps	Expected Result
Attempt to Create Account with Existing ID Number (Boundary Value Analysis)	Verify that the system does not allow the creation of multiple accounts using the same ID number.	1. The system does not allow duplicate ID numbers. 2. An account with ID:already exists	1. Create an account with ID: 123456, Password 2. Log out of the account. 3. Attempt to create another account using the same ID	Account creation fails, and the system displays an error message: "ID number already in use."

Requirement 4: The user shall be able to transfer funds between accounts whenever needed.

Test Case 4.1	Description	Preconditions	Test Steps	Expected Result
Transfer funds between accounts whenever needed successfully (Equivalence Partitioning)	Verify that the user can successfully transfer funds between two valid accounts.	1. The user has at least two accounts registered in the system. 2. Both accounts have a sufficient balance for the transfer.	1. Navigate to the "Transfer Funds" section. 2. Select the source account. 3. Select the destination account. 4. Enter a valid transfer amount. 5. Click the "Transfer" button.	The system confirms that the transfer was successful and updates the balances of both accounts accordingly.

Test Case 4.2	Description	Preconditions	Test Steps	Expected Result
Attempt to transfer funds between accounts without sufficient funds. (Boundary Value Analysis)	Verify that the user cannot transfer an amount exceeding the account balance.	The source account has a balance lower than the transfer amount.	1. Navigate to the "Transfer Funds" section. 2. Select the source account. 3. Select the destination account. 4. Enter an amount larger than the available balance. 5. Click the "Transfer" button.	The system displays an error message: "Insufficient funds for this transfer."

Requirement 5: The user shall be able to withdraw cash money from any ATM machine near them.

Test Case 5.1	Description	Preconditions	Test Steps	Expected Result
Make a cash withdrawal from an ATM successfully. (Equivalence Partitioning)	Verify that the user can withdraw cash successfully from an ATM with sufficient balance in their account.	The user is logged in and has an account with enough balance.	1. Go to the ATM withdrawal section. 2. Enter the withdrawal amount (less than the account balance). 3. Confirm the withdrawal.	The system processes the withdrawal, deducts the amount from the balance, and dispenses the cash.

Test Case 5.2	Description	Preconditions	Test Steps	Expected Result
Attempt to make a cash withdrawal from an ATM with insufficient funds. (Boundary Value Analysis)	Verify that the user cannot withdraw more cash than what is available in the account.	The user's account balance is lower than the requested withdrawal amount.	1. Go to the ATM withdrawal section. 2. Enter an amount larger than the available account balance. 3. Confirm the withdrawal.	The system displays an error message: "Insufficient funds."

Requirement 6: The user shall be able to deposit cash money at any ATM machine dedicated to deposit.

Test Case 6.1	Description	Preconditions	Test Steps	Expected Result
Deposit cash into account using ATM machine successfully. (Equivalence Partitioning)	Verify that the user can successfully deposit cash into their account.	The user is near an ATM that supports cash deposits. The cash deposited is not damaged.	1. Insert ATM card and enter the PIN. 2. Select the "Deposit Cash" option. 3. Insert the cash into the deposit slot. 4. Confirm the amount displayed.	The deposit is successful, and the amount is added to the account balance.

Test Case 6.2	Description	Preconditions	Test Steps	Expected Result
Attempt to deposit cash into account using ATM machine with damaged notes. (Error Guessing)	Verify that the system does not accept damaged notes.	The user attempts to deposit a damaged note.	1. Insert ATM card and enter the PIN. 2. Select the "Deposit Cash" option. 3. Insert damaged notes into the deposit slot. 4. Confirm the amount displayed.	The system rejects the deposit and returns the notes.

C: Automated testing

GITHUB CI/CD tool test:

We've done multiple tests on our codes using CI/CD tool, where all our tests passed as shown in the following picture

✓ CI CI #28: Manually run by Njoodo	main	36 minutes ago 15s	...
✓ Create workflowflutter_ci_cd.yml CI #27: Commit b8929b7 pushed by lujaintayy	main	49 minutes ago 11s	...
✓ Update flutter-analysis.yml CI #25: Commit 1df67a3 pushed by lujaintayy	main	1 hour ago 14s	...
✓ Update flutter-analysis.yml CI #24: Commit c021735 pushed by lujaintayy	main	1 hour ago 12s	...
✓ Update flutter-analysis.yml CI #23: Commit 2e3562c pushed by lujaintayy	main	1 hour ago 11s	...
✓ TEST withdrawal_action_model.dart CI #22: Commit cc444f3 pushed by lujaintayy	main	1 hour ago 14s	...
✓ Update issue templates CI #21: Commit 21b0317 pushed by lujaintayy	main	2 hours ago 10s	...
✓ Update flutter-analysis.yml CI #20: Commit bc49f57 pushed by lujaintayy	main	2 hours ago 15s	...

Activate Windows

More details about the tests

← CI

TEST withdrawal_action_model.dart #22

Re-run all jobs



Summary

Jobs

build

Run details

Usage

Workflow file

Triggered via push 1 hour ago

lujaintayy pushed cc444f3 main

Status

Success

Total duration

14s

Artifacts

-

blank.yml

on: push

build

4s



← CI

Update flutter-analysis.yml #23

Re-run all jobs



Summary

Jobs

build

Run details

Usage

Workflow file

Triggered via push 1 hour ago

lujaintayy pushed 2e3562c main

Status

Success

Total duration

11s

Artifacts

-

blank.yml

on: push

build

3s

